Notes: Hvide & Meling (RFS, 2023)

Do Temporary Demand Shocks Have Long-Term Effects for Startups?

Contents

1 Big picture

- **Question.** Can a temporary demand shock permanently change the trajectory of a startup?
- **Empirical setting.** The paper uses Norwegian public procurement auctions run by the Norwegian Public Roads Administration. The key comparison is between startups that win an auction and startups that narrowly lose, usually by coming second.
- **Treatment.** Winning a procurement auction creates a temporary demand shock: the startup gets a short-term contract, usually less than one year, whose average value is large relative to the startup's pre-auction sales.
- **Main result.** Startup winners are substantially larger than runner-up startups even several years after the contract work has ended. In the post-contract period, winners have about 24% higher sales and about 22% more employees.
- **Economic magnitude.** The estimated long-run sales increase is roughly 50% of the contract value, implying that a short-lived demand shock transmits into persistent firm growth.
- **Startup-specific effect.** The effect is strong for startups but disappears for mature firms, consistent with decreasing returns to experience or learning.
- **Identification strategy.** The baseline design is a difference-in-differences event study comparing first-time auction winners with runner-up startups. Firm-by-auction fixed effects absorb time-invariant startup quality; event-time and calendar-year effects absorb common timing patterns.
- **Key identification checks.** Winners and runners-up are balanced before the auction; runner-up versus third-place placebo comparisons show no post-auction difference; timing-of-event, within-auction, and regression-discontinuity analyses yield similar estimates.
- **Mechanism evidence.** Winners later participate in larger auctions, face stronger competitors, enter new product categories, and hire higher-quality managers. Startups bid more aggressively than mature firms. These patterns support learning-by-doing.
- **Alternative mechanisms.** Sunk investments, contract delays, cost overruns, financial constraints, and reputation are considered. Some can contribute, but the broad pattern fits learning-by-doing best.
- **Takeaway.** Startup growth is path dependent. Temporary demand shocks can create durable capability gains and long-run differences in size, especially for young firms with low accumulated experience.

2 Institutional background and data

2.1 Norway and Public Roads procurement

- Norway provides unusually rich administrative data linking firm accounts, ownership, employment, wages, and procurement participation.
- The procurement authority is the Norwegian Public Roads Administration, responsible for road construction, maintenance, and many related purchases.
- Public Roads uses competitive procurement auctions for road work, consulting, supplies, maintenance, safety measures, and related services.
- Auctions are typically short-term, with contract duration often below one year.
- The paper focuses mainly on price-only procurement auctions, where the lowest bid wins. This makes the ranking of winner, runner-up, and third-place bidder especially transparent.

Interpretation: The setting is attractive because procurement wins create large but temporary demand shocks, while administrative data make it possible to track firm outcomes for many years after the contract.

2.2 Register data

- The authors use Norwegian register data covering all Norwegian firms.
- These data include accounting variables, employment, wages, ownership links, and firm age.
- The data allow the authors to distinguish startups from subsidiaries or spinoffs of established companies.
- Key firm outcomes include sales, value added, employment, profits, wage bill, intermediate inputs, tangible assets, total assets, debt, paid-in capital, leverage, and worker/CEO quality.

Interpretation: The administrative data make the outcome measurement unusually broad. The paper is not restricted to survival or revenue; it can measure inputs, assets, human capital, product-market choices, and future auction behavior.

2.3 Procurement-level data

- Public Roads provides transcripts for procurement auctions between 2003 and 2015.
- For each auction, the data include the product or service procured, contract duration, the identity of each bidder, and bid size.
- The paper classifies procurements into 54 product categories based on detailed text descriptions.
- Figure 1 shows that procurement categories include road work, traffic safety, pedestrian/bicycle paths, bridge work, noise cancellation, demolition, operation and maintenance, tunnel work, signs, and many smaller categories.

Interpretation: Product categories matter because the mechanism tests ask whether winners expand vertically into larger/more demanding auctions and horizontally into new product markets.

2.4 Startup definition and estimation sample

- The baseline sample consists of startup-by-auction observations.
- Startups are young firms, not subsidiaries of established companies.
- The treated group consists of first-time winners.
- The control group consists of runner-up startups that have not previously won an auction.
- The main estimation sample in Table 1 has 361 startup-by-auction observations.
- In the year before the focal auction, the average startup has 17.2 employees, NOK 24.2 million in sales, NOK 7.3 million in total wage bill, and NOK 13.4 million in total assets.
- The mean estimated contract value is NOK 12.2 million, and the mean bid size is NOK 9.6 million.

Interpretation: The contract is economically large relative to startup size. The average contract is roughly one-half of prior-year sales, so winning is a meaningful temporary demand shock.

3 Empirical methodology

3.1 Baseline difference-in-differences model

The unit of observation is startup-by-auction j in event time e. The core model is

$$y_{je} = \theta b_{je} + \kappa_t + \alpha_e + \lambda_j + \varepsilon_{je}. \tag{1}$$

- y_{je} is an outcome such as log sales, log value added, log employment, profits, or log inputs.
- $b_{je} = Treated_j \times Post_e$.
- $Treated_j = 1$ for startups that win the auction and zero for runners-up.
- $Post_e = 1$ in years after the auction and zero before.
- κ_t are calendar-year fixed effects.
- α_e are event-time fixed effects.
- λ_j are firm-by-auction fixed effects.
- Standard errors are clustered at the firm level.

Interpretation: The coefficient θ is a within-startup before-after difference for winners relative to runners-up. It is interpreted as the causal effect of winning if winners and runners-up would have followed parallel trends absent the win.

3.2 Contract and post-contract effects

The paper lets θ differ across two periods:

- **Contract period:** years in which the procurement contract is being performed.
- **Post-contract period:** years after the contract work has ended.

Interpretation: This distinction is crucial. A temporary demand shock mechanically raises outcomes during the contract period. The paper's central claim is that effects persist after the contract is over.

3.3 Why compare winners with runner-ups?

- Runner-ups are likely to be more comparable to winners than all losing bidders.
- Both winners and runner-ups chose to bid and were competitive.
- In price-only auctions, the winner and runner-up are separated by a narrow bid-ranking margin.
- The design also excludes previous winners from the control group to avoid contaminating controls with earlier treatment effects.

Interpretation: The empirical strategy is built to compare firms that almost received the same temporary demand shock.

3.4 Alternative bid-level specification

For the mechanism test comparing startup and mature firms' bids, the paper estimates

$$Y_{ik} = \alpha_k + \beta Startup_{ik} + \epsilon_{ik}. \tag{2}$$

- Y_{ik} is log bid size, or an indicator for winning, for bidder i in auction k.
- $Startup_{ik}$ indicates that the bidder is a startup rather than a mature firm.
- α_k are auction fixed effects.

Interpretation: Auction fixed effects mean startups are compared with mature firms bidding in the exact same auction. A negative β for log bids means startups bid more aggressively.

4 Identification and credibility

4.1 Common trends evidence

- Figure 2 compares pre-auction levels and trends for winners versus runners-up and runner-ups versus third-place startups.
- Winners and runners-up are balanced in pre-auction levels and trends for major firm characteristics.
- Runner-ups and third-place bidders also look similar, providing a placebo comparison.
- Pre-auction event-study coefficients for value added and employment show no meaningful placebo effects.

Interpretation: The identifying assumption is not directly testable, but the pre-trend evidence strongly supports the idea that winners and runner-ups are comparable before treatment.

4.2 Placebo tests

- The first placebo compares runner-ups to third-place bidders. Neither group wins, so any post-auction difference would signal a ranking-related confound.
- The paper finds no meaningful post-auction difference between runner-ups and third-place startups.
- This helps rule out the concern that simply being closer to winning predicts future growth.

Interpretation: If runner-ups and third-place firms evolve similarly, the winner-runner-up gap is less likely to be driven by unobserved quality differences among bidders.

4.3 Contract delays and cost overruns

- Contract delays could contaminate post-contract estimates if work spills into later years.
- Table 3 shows the mean delay is only 1.52 months in the full sample and 1.76 months in the matched startup sample.
- Median delays are 0 months in the full sample and 1 month in the matched startup sample.
- The 95th percentile delay is 8 months in the full sample and 6 months in the matched startup sample.
- Cost overruns are common but do not explain persistent effects several years after completion.

Interpretation: Contract delays are too short to explain the long-term post-contract effects in Figure 2 and Table 2.

4.4 Additional identification checks

- **Timing-of-event specification:** compares current winners to startups that win in future years.
- **Within-auction specification:** compares winners and runner-ups from the same auction.
- **Regression discontinuity design:** compares winners and losers in narrowly won auctions.
- These checks produce effects similar to the baseline difference-in-differences estimates.

Interpretation: The result is robust to multiple designs that use different identifying variation. This reduces concern that any one modeling choice drives the findings.

5 Main results and mechanisms

5.1 Main treatment effects

- Table 2 shows that winners grow strongly during the contract period and remain larger after contract completion.
- In the contract period, winners have 25% higher sales, 19% higher value added, 14% higher employment, and higher profits.
- In the post-contract period, winners have 24% higher sales, 19% higher value added, 22% higher employment, and higher profits.
- Winners do not have lower survival; the active-firm indicator is essentially unchanged.
- Winners increase intermediate inputs during the contract period and tangible assets after the contract period.

Interpretation: The shock is not simply a short-lived sales spike. The winning startups become persistently bigger and more profitable.

5.2 Economic magnitude

- The paper's back-of-the-envelope calculation uses a 24% long-run sales increase from a base of NOK 24 million.
- This implies a long-run sales increase of about NOK 5.76 million.
- Relative to a mean contract value of NOK 11 million, about 50% of the contract value is transmitted into long-run sales.

Interpretation: The long-run effect is economically large. It is not merely statistically significant.

5.3 Treatment intensity and firm age

- Figure 3 shows that effects are larger when the contract is large relative to the startup's pre-auction sales.
- Figure 4 shows that long-run effects are strongest for young firms and disappear for mature firms.
- Effects are statistically and economically small for firms aged above 10 years.

Interpretation: These patterns fit learning-by-doing. A given experience shock matters more when accumulated experience is low and when the shock is large relative to the firm's current scale.

5.4 Startup dynamics after winning

- Winners increase paid-in capital and reduce leverage.
- Winners hire higher-quality CEOs and pay CEOs more in the post-contract period.
- Winners are more likely to participate in future Public Roads auctions.
- Winners are more likely to compete in larger auctions, against higher-TFP competitors, and in new product categories.
- Winners are also more likely to win auctions for new products.

Interpretation: The firm evolves after winning. The evidence points to capability building, not merely passive receipt of revenue.

5.5 Learning-by-doing mechanism

- Learning-by-doing predicts that experience improves capabilities and generates persistent differences in productivity or market reach.
- It also predicts decreasing returns: young firms learn more at the margin than mature firms.
- The age-bin evidence, vertical/horizontal expansion, and aggressive startup bidding all support this mechanism.

Interpretation: Learning-by-doing is the paper's preferred explanation because it explains why effects are persistent, startup-specific, and stronger for larger treatment intensity.

5.6 Alternative mechanisms

- **Sunk investments:** winners may invest during the contract period, lowering future costs. But tangible assets rise mainly after the contract period, suggesting investment is more consequence than cause.
- **Financial constraints:** winners expand intermediate inputs, but effects do not vary strongly with conventional financial constraint measures.
- **Reputation and demand frictions:** winning may help firms become known. Newspaper case evidence shows mentions of learning, network, and reputation mechanisms.
- **Contract delays:** delays are too short to explain the long-run effects.

Interpretation: Several channels may interact, but the broad pattern points most strongly to learning-by-doing and capability accumulation.

6 Tables and figures

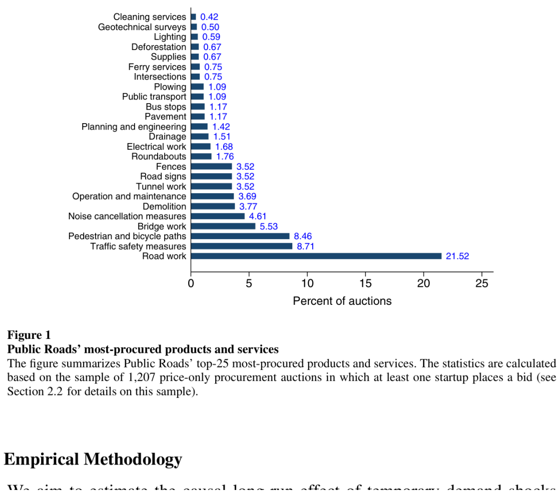
Visuals are directly extracted from the paper and grouped to improve reading speed.

6.1 Figure 1 and Table 1: procurement products and estimation sample

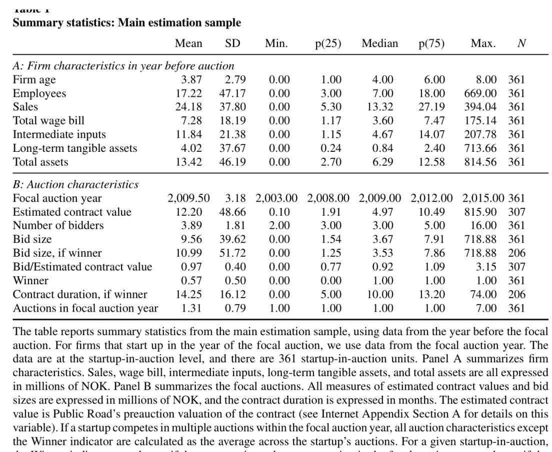
Read:

- Figure 1 shows the top 25 products and services procured by Public Roads in auctions where startups bid.
- Road work is the largest category, accounting for more than 20% of auctions.
- Other frequent categories include traffic safety measures, pedestrian and bicycle paths, bridge work, noise cancellation, demolition, operation and maintenance, tunnel work, road signs, fences, and roundabouts.
- Table 1 shows the startups are small and young before the focal auction: mean age 3.87 years, 17.22 employees, NOK 24.18 million in sales, and NOK 13.42 million in assets.
- The mean estimated contract value is NOK 12.20 million and the mean bid size is NOK 9.56 million.

Economic message: The treatment is a large demand shock for small firms, and the procurement categories are broad enough to capture heterogeneous production experiences.



(a) Figure 1: Public Roads' most-procured products and services.



Summary statistics: Main estimation sample

	Mean	SD	Min.	p(25)	Median	p(75)	Max.	*N*
A: Firm characteristics in year before auction								
Firm age	3.87	2.79	0.00	1.00	4.00	6.00	8.00	361
Employees	17.22	47.17	0.00	3.00	7.00	18.00	669.00	361
Sales	24.18	37.80	0.00	5.30	13.32	27.19	394.04	361
Total wage bill	7.28	18.19	0.00	1.17	3.60	7.47	175.14	361
Intermediate inputs	11.84	21.38	0.00	1.15	4.67	14.07	207.78	361
Long-term tangible assets	4.02	37.67	0.00	0.24	0.84	2.40	713.66	361
Total assets	13.42	46.19	0.00	2.70	6.29	12.58	814.56	361
B: Auction characteristics								
Focal auction year	2,009.50	3.18	2,003.00	2,008.00	2,009.00	2,012.00	2,015.00	361
Estimated contract value	12.20	48.66	0.10	1.91	4.97	10.49	815.90	307
Number of bidders	3.89	1.81	2.00	3.00	3.00	5.00	16.00	361
Bid size	9.56	39.62	0.00	1.54	3.67	7.91	718.88	361
Bid size, if winner	10.99	51.72	0.00	1.25	3.53	7.86	718.88	206
Bid/Estimated contract value	0.97	0.40	0.00	0.77	0.92	1.09	3.15	307
Winner	0.57	0.50	0.00	0.00	1.00	1.00	1.00	361
Contract duration, if winner	14.25	16.12	0.00	5.00	10.00	13.20	74.00	206
Auctions in focal auction year	1.31	0.79	1.00	1.00	1.00	1.00	7.00	361

The table reports summary statistics from the main estimation sample, using data from the year before the focal auction. For firms that start up in the year of the focal auction, we use data from the focal auction year. The data are at the startup-in-auction level, and there are 361 startup-in-auction units. Panel A summarizes firm characteristics. Sales, wage bill, intermediate inputs, long-term tangible assets, and total assets are all expressed in millions of NOK. Panel B summarizes the focal auctions. All measures of estimated contract values and bid sizes are expressed in millions of NOK, and the contract duration is expressed in months. The estimated contract value is Public Road's preauction valuation of the contract (see Internet Appendix Section A for details on this variable). If a startup competes in multiple auctions within the focal auction year, all auction characteristics except the Winner indicator are calculated as the average across the startup's auctions. For a given startup-in-auction,

(b) Table 1: Summary statistics.

6.2 Figure 2 and Table 2: identification and main effects

Read:

- Figure 2 shows pre-auction balance and event-time effects. Winners and runners-up are similar before the auction, and runner-ups and third-place firms show no placebo divergence.
- Table 2 reports the baseline difference-in-differences estimates.
- During the contract period, sales rise by 25%, value added by 19%, employment by 14%, profits by NOK 583 thousand, and intermediate inputs by 34%.
- In the post-contract period, sales rise by 24%, value added by 19%, employment by 22%, profits by NOK 813 thousand, wage bill by 23%, intermediate inputs by 22%, and tangible assets by 38%.
- The active-firm indicator does not change, indicating the results are not driven by differential survival.

Economic message: Winning creates persistent firm growth. The post-contract effects are the core evidence of long-run path dependence from temporary demand shocks.

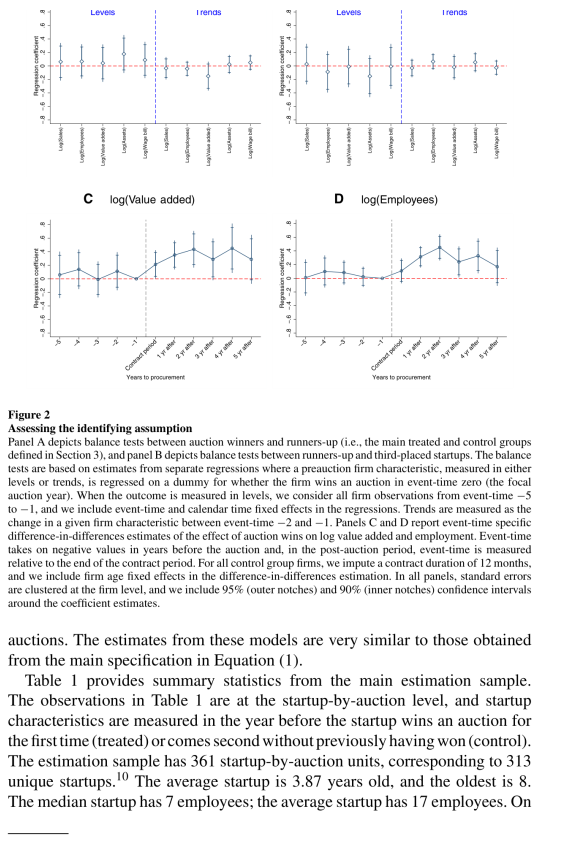


Figure 2
Assessing the identifying assumption
Panel A depicts balance tests between auction winners and runners-up (i.e., the main treated and control groups defined in Section 3), and panel B depicts balance tests between runners-up and third-placed startups. The balance tests are based on estimates from separate regressions where a preauction firm characteristic, measured in either levels or trends, is regressed on a dummy for whether the firm wins an auction in event-time zero (the focal auction year). When the outcome is measured in levels, we consider all firm observations from event-time −5 to −1, and we include event-time and calendar time fixed effects in the regressions. Trends are measured as the change in a given firm characteristic between event-time −2 and −1. Panels C and D report event-time specific difference-in-differences estimates of the effect of auction wins on log value added and employment. Event-time takes on negative values in years before the auction and, in the post-auction period, event-time is measured relative to the end of the contract period. For all control group firms, we impute a contract duration of 12 months, and we include firm age fixed effects in the difference-in-differences estimation. In all panels, standard errors are clustered at the firm level, and we include 95% (outer notches) and 90% (inner notches) confidence intervals around the coefficient estimates.

auctions. The estimates from these models are very similar to those obtained from the main specification in Equation (1).

Table 1 provides summary statistics from the main estimation sample. The observations in Table 1 are at the startup-by-auction level, and startup characteristics are measured in the year before the startup wins an auction for the first time (treated) or comes second without previously having won (control). The estimation sample has 361 startup-by-auction units, corresponding to 313 unique startups.[10] The average startup is 3.87 years old, and the oldest is 8. The median startup has 7 employees; the average startup has 17 employees. On

(a) Figure 2: Balance and event-time effects.

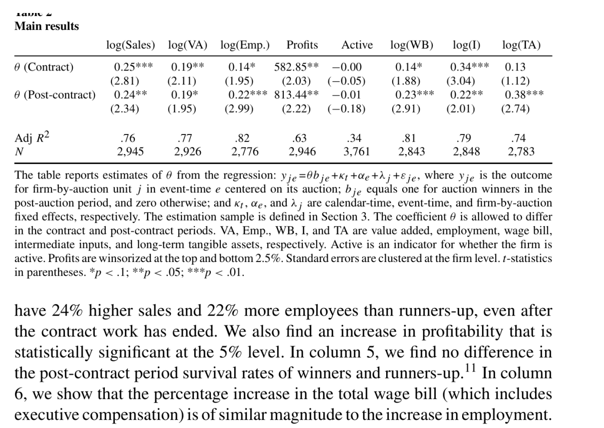


Table 2
Main results

	log(Sales)	log(VA)	log(Emp.)	Profits	Active	log(WB)	log(I)	log(TA)
θ (Contract)	0.25***	0.19**	0.14*	582.85**	−0.00	0.14*	0.34***	0.13
	(2.81)	(2.11)	(1.95)	(2.03)	(−0.05)	(1.88)	(3.04)	(1.12)
θ (Post-contract)	0.24**	0.19*	0.22***	813.44**	−0.01	0.23***	0.22**	0.38***
	(2.34)	(1.95)	(2.99)	(2.22)	(−0.18)	(2.91)	(2.01)	(2.74)
Adj R^2	.76	.77	.82	.63	.34	.81	.79	.74
N	2,945	2,926	2,776	2,946	3,761	2,843	2,848	2,783

The table reports estimates of θ from the regression: $y_{je}=\theta b_{je}+\kappa_t+\alpha_e+\lambda_j+\varepsilon_{je}$, where y_{je} is the outcome for firm-by-auction unit j in event-time e centered on its auction; b_{je} equals one for auction winners in the post-auction period, and zero otherwise; and κ_t, α_e, and λ_j are calendar-time, event-time, and firm-by-auction fixed effects, respectively. The estimation sample is defined in Section 3. The coefficient θ is allowed to differ in the contract and post-contract periods. VA, Emp., WB, I, and TA are value added, employment, wage bill, intermediate inputs, and long-term tangible assets, respectively. Active is an indicator for whether the firm is active. Profits are winsorized at the top and bottom 2.5%. Standard errors are clustered at the firm level. t-statistics in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

have 24% higher sales and 22% more employees than runners-up, even after the contract work has ended. We also find an increase in profitability that is statistically significant at the 5% level. In column 5, we find no difference in the post-contract period survival rates of winners and runners-up.[11] In column 6, we show that the percentage increase in the total wage bill (which includes executive compensation) is of similar magnitude to the increase in employment.

(b) Table 2: Main results.

6.3 Figure 3 and Figure 4: treatment intensity and age heterogeneity

Read:

- Figure 3 plots effects by treatment intensity, defined by contract size relative to firm scale.
- Effects on value added and employment increase with treatment intensity.
- Figure 4 plots effects by firm age.
- Effects are strongest for the youngest startups and become insignificant for firms above 10 years old.

Economic message: A larger experience shock and lower accumulated experience generate stronger long-run effects. This is exactly what learning-by-doing predicts.

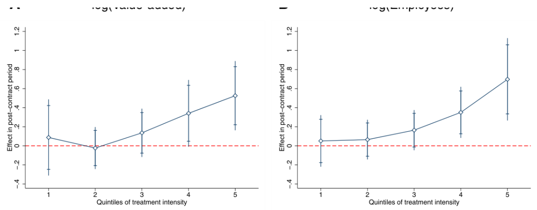
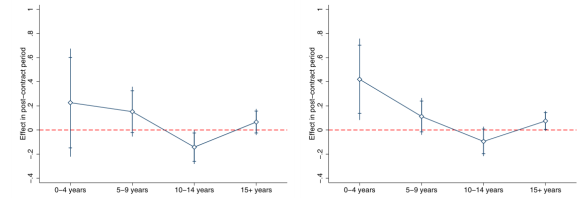


Figure 3
Effect on value added and employment by treatment intensity
The figure reports estimates of the post-contract treatment effect of procurement auction wins on log(Value added) and log(Employees). The post-contract treatment effect is estimated for quintiles of treatment intensity using the following regression specification: $y_{je}=\theta b_{je}\times \text{Quint}_j+\kappa_t+\alpha_e+\lambda_j+\varepsilon_{je}$, where y_{je} is the outcome for firm-by-auction j in event-time e centered on its auction; Quint_j is a vector of indicators for treatment intensity quintile; b_{je} equals 1 for auction winners in the post-auction period, 0 otherwise; and κ_t, α_e, and λ_j are calendar-time, event-time, and firm-by-auction fixed effects, respectively. The estimation sample is described in Section 3. Treatment intensity is measured as the total auction winnings in event-time 0 divided by sales in event-time −1. The coefficient θ is allowed to differ in the procurement contract period and post-contract period, and we only report the estimates of θ from the post-contract period. Standard errors are clustered at the firm level. The figure includes 95% (outer notches) and 90% (inner notches) confidence intervals around the coefficient estimates.

for winners and runners-up over event-time, where event-time is defined relative to the end of the contract duration (e.g., "2yr after" means 2 years after contract

(a) Figure 3: Effects by treatment intensity.



igure 4
Effect on value added and employment by age bin
The figure reports the firm-age-specific post-contract treatment effect of procurement auction wins on log(Value added) and log(Employees). The treatment effects are estimated separately for firms aged 0–4, 5–9, 10–14, and 15+ years in the focal auction year. Specifically, we estimate the following regression specification within each age bin: $y_{je}=\theta b_{je}+\kappa_t+\alpha_e+\lambda_j+\varepsilon_{je}$, where y_{je} is the outcome for firm-by-auction unit j in event-time e centered on its auction; b_{je} equals one for auction winners in the post-auction period, zero otherwise; and κ_t, α_e, and λ_j are calendar-time, event-time, and firm-by-auction fixed effects, respectively. The coefficient θ is allowed to differ in the procurement contract and post-contract periods, and we only report the estimates of θ from the post-contract period. The estimation sample comprises both startup and nonstartup auction winners and runner-ups. Standard errors are clustered at the firm level, and the figure includes 95% (outer notches) and 90% (inner notches) confidence intervals around the coefficient estimates.

and estimate the effects of wins for these "pseudo-startups". We find no long-run

(b) Figure 4: Effects by age bin.

6.4 Table 3: contract delays and cost overruns

Read:

- In the full contract sample, the mean delay is 1.52 months and the median delay is 0 months.
- In the matched startup sample, the mean delay is 1.76 months and the median delay is 1 month.
- The 95th percentile delay is only 8 months in the full sample and 6 months in the matched sample.
- Cost overruns are common, with mean overruns of 15.58% in the full sample and 12.90% in the matched sample.

Economic message: Delays and overruns exist, but delays are too short to explain persistent multi-year post-contract growth.

Table 3
Contract delays and cost overruns

	Mean	p(5)	p(10)	p(25)	p(50)	p(75)	p(90)	p(95)	*N*
A. Full sample									
Delay (months)	1.52	−1.00	0.00	0.00	0.00	2.00	6.00	8.00	1,749
Cost overrun (%)	15.58	−18.89	−10.85	−0.59	10.27	26.47	50.45	72.57	1,749
B. Matched sample									
Delay (months)	1.76	0.00	0.00	0.00	1.00	3.00	5.00	6.00	45
Cost overrun (%)	12.90	−19.96	−8.86	−2.65	7.21	18.79	46.28	58.99	45

The table reports summary statistics on contract delays and cost overruns relative to the winning auction bid. Panel A summarizes the full sample of 1,749 contracts. Panel B summarizes a subsample of 45 contracts won by startups. Internet Appendix Section E provides more details on the data and sample construction.

come second, even several years after the contract work ends. In this section, we investigate whether these long-term effects can be interpreted as causal.

Table 4
Other outcomes

A: Capital structure and employee quality

	log(Debt)	log(Paid-in cap.)	Lev.	CEO quality$^{\text{Switchers}}$	CEO quality$^{\text{All}}$	log(CEO wage)	Worker quality	log(Avg. worker wage)
θ (Contract)	0.11	0.18**	−0.06**	0.10	0.05*	0.08	−0.02	−0.02
	(1.37)	(2.24)	(−2.07)	(1.00)	(1.74)	(1.00)	(−1.01)	(−0.95)
θ (Post-contract)	0.08	0.28**	−0.08**	0.13	0.06	0.14**	−0.02	−0.01
	(0.87)	(2.55)	(−2.38)	(0.90)	(1.58)	(2.22)	(−0.89)	(−0.27)
Adj R^2	.78	.82	.47	.72	.89	.59	.76	.78
N	2,942	2,891	2,944	351	1,604	1,633	2,263	2,270

B: Later auction participation and winning

	Participation						Winning	
	Competes	Large	Larger	High TFP	New product	Quality	New product	Quality
θ (Contract)	0.05*	0.10***	0.01	0.09**	0.11***	0.00	0.05***	−0.00
	(1.76)	(2.76)	(0.66)	(2.53)	(2.89)	(0.02)	(3.11)	(−0.06)
θ (Post-contract)	0.13***	0.07***	0.12***	0.12***	0.10***	0.01	0.05**	0.01
	(3.74)	(3.02)	(4.21)	(4.10)	(3.11)	(0.59)	(2.56)	(0.66)
Adj R^2	.53	.32	.28	.35	.30	.22	.19	.14
N	3,761	3,761	3,761	3,606	3,761	3,761	3,761	3,761

The table reports estimates of θ from the following regression: $y_{je}=\theta b_{je}+\kappa_t+\alpha_e+\lambda_j+\varepsilon_{je}$, where y_{je} is the outcome for firm-by-auction unit j in event-time e centered on its auction; b_{je} equals one for auction winners in the post-auction period, zero otherwise; and κ_t, α_e, and λ_j are calendar-time, event-time, and firm-by-auction fixed effects, respectively. The estimation sample is defined in Section 3. The coefficient θ is allowed to differ in the contract and post-contract periods. In panel A, CEO and worker quality are measured as the CEO's or worker's wage in the year before the focal auction, as explained in Section 6.1. In columns 4–6, we include in the estimation sample only the firms that have a CEO both before and after the focal auction. In column 4, we further restrict attention to firms that change CEOs in the post-auction period. In panel B, the outcomes are indicators for whether a firm competes in a price-only auction; competes in a large (above median value) auction; competes in an auction that is larger than the focal auction; competes in auctions in which the competing bidders are above-median TFP; competes in auctions for new products; competes in auctions in which price is not the only winning criterion; wins in auctions for new products; or wins in auctions in which price is not the only winning criterion. Standard errors are clustered at the firm level. t-statistics in parentheses. $^*p<.1$; $^{**}p<.05$; $^{***}p<.01$.

6.5 Table 4: financing, restaffing, and product-market choices

Read:

- Panel A shows winners increase paid-in capital and reduce leverage.
- CEO quality increases modestly, and CEO wage rises in the post-contract period.
- Worker quality and average worker wage do not change much.

- Panel B shows winners become more likely to compete in future auctions, large auctions, auctions larger than the focal auction, auctions with high-TFP competitors, and auctions for new products.
- Winners also become more likely to win auctions for new products.

Economic message: Auction winners do more than scale up. They move into more demanding and broader markets, suggesting capability accumulation and learning.

6.6 Table 5 and Figure 5: bidding and case evidence

Read:

- Table 5 shows startups bid 1.4% lower than mature firms in the same auction.
- Startups also have a 2.5 percentage point higher probability of winning.
- Younger startups appear to bid more aggressively than older startups.
- Figure 5 summarizes newspaper case evidence. Larger contracts are more likely to be found in newspaper cases.
- Most newspaper mentions involve interviews with startup founders or CEOs.
- Mechanism mentions include learning, reputation, networks, and investment-related narratives.

Economic message: Startups seem to recognize that winning has future value, and case evidence supports mechanisms in which contracts build capabilities and market access.

Table 5
Comparing startup and mature firms' auction bids

	log(Bid)	Winner	log(Bid)
Startup	−0.014**	0.025*	
	(−2.399)	(1.651)	
Startup & Age < 5			−0.017*
			(−1.904)
Startup & Age ≥ 5			−0.012
			(−1.586)
Adj R^2	.99	−.26	.99
N	12,729	12,729	12,729

The first two columns report estimates of β from Equation (2) using log bid size and an auction win indicator as the outcomes. The data are at the bid level, and there are 12,729 bids spread out across 3,677 Public Roads price-only auctions (Table IA.14 in the Internet Appendix provides summary statistics). The third column reports estimates from a modified version of Equation (2) in which Startup_i has been replaced with separate indicators for startups aged below and above 5 years. Standard errors are clustered at the firm level. t-statistics in parentheses. t-statistics in parentheses. $*p < .1$; $**p < .05$; $***p < .01$.

(a) Table 5: Startup versus mature-firm bids.

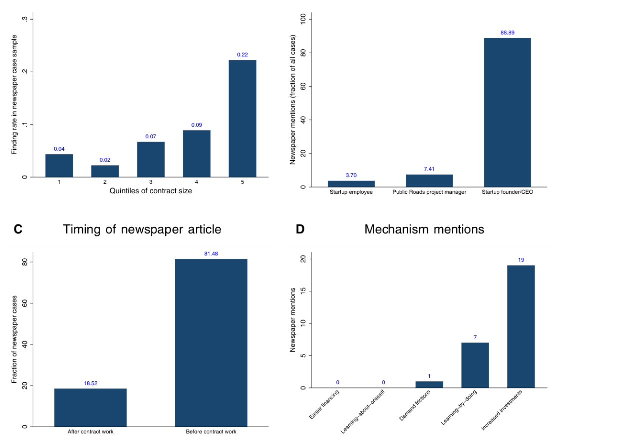


Figure 5
Summary statistics: Newspaper cases
The figure presents summary statistics from 27 newspaper articles describing startups' experiences with winning Public Roads procurement auctions. In panel A, we plot the "finding rate" (the count of firms in the newspaper sample divided by the count of startup winners in the estimation sample) by quintiles of Public Roads contract size. Contract size is measured as the contract value divided by firm sales in the year before the auction. Panel B describes the interviewees in the 27 newspaper articles. The bars in panel B sum to 100% because all of the sampled newspaper articles include an interview. Panel C describes whether the newspaper articles were written before or after the Public Roads contract work had started. Panel D counts the number of newspaper articles that mention each of the five main theoretical mechanisms (investment, learning-by-doing, demand frictions, learning about oneself, and relaxed financial constraints) discussed in Section 7.2.

into product areas that are new to the startup. Thus, with a broad definition of learning, 7 (26%) cases are included. Regarding *demand frictions*, only one (4%) case mentions that the contract experience can be used as a reference to attract new customers later on. None of the newspaper articles mentions that the contract will ease *financial constraints* (some cases describe the need to

(b) Figure 5: Newspaper case evidence.

7 Short summary

- The paper studies whether temporary demand shocks have long-term effects on startups.
- The empirical setting is Norwegian public procurement auctions.
- The treatment is winning a procurement contract; the control group is runner-up startups.
- The baseline model is a difference-in-differences event study with calendar-year, event-time, and firm-by-auction fixed effects.
- Winners and runners-up are balanced before the auction.
- Runner-up versus third-place placebo tests show no comparable post-auction divergence.
- Winners have about 24% higher sales and 22% higher employment after the contract work ends.
- About 50% of contract value is transmitted into long-run sales.
- Effects are stronger for larger treatment intensity and younger firms.
- Mature firms show little or no long-run effect.
- Winners later compete in larger auctions and in new product categories.
- Startups bid more aggressively than mature firms, consistent with internalizing the future value of learning.
- Contract delays are too short to explain the results.
- Learning-by-doing is the preferred mechanism, though reputation and financial constraints may also matter.

8 Conclusion

8.1 Core conclusion

Hvide and Meling show that temporary demand shocks can create long-lasting differences in startup outcomes. Startups that win procurement auctions become persistently larger and more profitable than runner-up startups, even after the contract work has ended.

8.2 Economic intuition

A procurement contract gives a young firm experience, revenue, and a chance to solve production problems at a larger scale. For startups with little accumulated experience, this can trigger learning-by-doing, improved capabilities, and entry into more demanding product markets. The resulting growth can persist long after the original contract is complete.

8.3 Takeaway

The paper provides strong evidence of path dependence in startup growth. Early demand shocks can matter not only because they temporarily raise sales, but because they can alter the firm's capability set and future opportunity set.